🛵 Food Delivery Time Prediction

📌 Objective

This project aims to **predict food delivery times** based on multiple features like customer and restaurant location, weather conditions, traffic, vehicle type, and more. It covers data preprocessing, exploratory data analysis (EDA), feature engineering, and predictive modeling using **Linear Regression** and **Logistic Regression**.

📁 Dataset

File: Food_Delivery_Time_Prediction.csv
The dataset includes:

- Distance between restaurant and customer
- Delivery Time
- Weather and Traffic Conditions
- Order Cost
- Vehicle Type
- Delivery Person Experience
- Order Priority

🔎 Phase 1: Data Preprocessing & EDA

✅ Step 1: Data Cleaning & Encoding

- Handled missing values in key columns (Distance, Delivery_Time, etc.).
- Encoded categorical features using Label Encoding and One-Hot Encoding.
- Normalized continuous features (Distance, Order_Cost, Delivery_sTime).

📊 Step 2: Exploratory Data Analysis

- Performed descriptive statistics (mean, median, variance, etc.).
- Analyzed correlations with Delivery_Time.
- Detected outliers using boxplots and visual inspections.

🛠️ Step 3: Feature Engineering

- Calculated distance using Haversine formula (if coordinates were provided).
- Created time-related features like Rush_Hour to enhance model accuracy.

🤖 Phase 2: Predictive Modeling

📈 Linear Regression

- Split data (80% training, 20% testing).
- Used Linear Regression to predict exact delivery times.
- **Evaluation Metrics:**

- R-squared (R^2)
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)

⚖️ Logistic Regression

- Classified deliveries into "Fast" or "Delayed".
- Trained a Logistic Regression model using traffic, weather, and experience as features.
- **Evaluation Metrics:**
  - Accuracy
  - Precision & Recall
  - F1-score
  - Confusion Matrix

📊 Model Comparison & Insights

- Compared models using confusion matrices, ROC curves, and regression metrics.
- Visualized findings using:
  - Pairplots
  - Heatmaps
  - Boxplots
  - Confusion Matrices

💡 Actionable Insights

- Optimize delivery routes based on peak traffic hours.
- Increase staffing during high-demand times.
- Provide additional training to improve delivery efficiency.

🚀 Final Deliverables

File/Folder	Description
notebooks/	Jupyter notebook with full code & analysis
data/	Dataset file (.csv)
images/	Visualizations (optional)
README.md	Project overview and documentation
requirements.txt	(Optional) Python dependencies

🛠 Tools & Libraries Used

- Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn)
- Jupyter Notebook
- Scikit-learn for modeling
- Git & GitHub for version control